

INFORMATION TECHNOLOGY INSTITUTE

LAB 5

Simple vs Wide Swing Cascode Current Mirror





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Part 1: Exploring Sizing Tradeoffs Using SA

Sinking current means which device type? NMOS or PMOS?

Sinking current means NMOS device

The Change in current translates to a spec on the λ of the device and required λ .

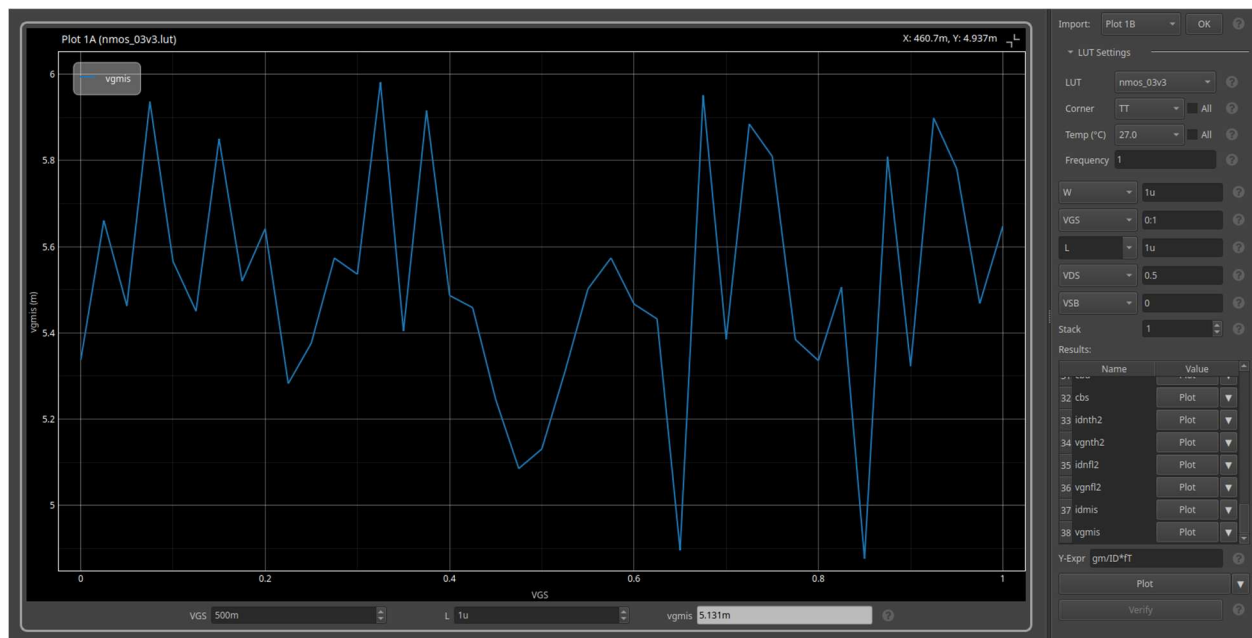
$$I_D = \frac{1}{2} k V_{OV}^2 (1 + \lambda V_{DS})$$

So the change in I_D is $\Delta I_D = \frac{1}{2} k V_{OV}^2 \lambda V_{DS}$ so $\frac{\Delta I_D}{I_{D0}} = \lambda V_{DS}$ Then $\lambda V_{DS} < 0.1$

$$\lambda < 0.1 \quad \text{at } V_{DS} = 1V$$

$$\lambda < 0.2 \quad \text{at } V_{DS} = 0.5V$$

VGMIS From SA





Average is around **5.5mV** for a unity area MOSfet

σ_{VT} from PDK files

```

46930 .lib fets_mn
46931 .subckt nfet_03v3 d g s b w=1e-5 l=2.8e-7 nf=1 m=1 as=0 ad=0 ps=0 pd=0 nrd=0 nrs=0 par=1 dtemp=0 sa=0 sb=0 sd=0
46932 .param par_vth=0.007148
46933 .param par_k=0.007088
46934 .param par_l=1.5e-7
46935 .param par_w=-1e-7
46936 .param par_leff='l-par_l'
46937 .param par_weff='par*(w-par_w)'
46938 .param p_sqrtarea='sqrt((par_leff)*(par_weff))'
46939 .param var_k='0.7071*par_k*1e-06/p_sqrtarea'
46940 .param mis_k='agauss(0,var_k,1)'
46941 .param var_vth='0.7071*par_vth*1e-06/p_sqrtarea'
46942 .param mis_vth='agauss(0,var_vth,1)'
46943 m0 d g s b nfet_03v3 ad=ad as=as delvto='mis_vth*sw_stat_mismatch' l=l m=m mulu0='1-mis_k*sw_stat_mismatch' nf=nf nrd=nrd nrs=nrs pd=pd ps=ps sa=sa sb=sb sd=sd w=w
46944 .ends nfet_03v3
46945 .subckt pfet_03v3 d g s b w=1e-5 l=2.8e-7 nf=1 m=1 as=0 ad=0 ps=0 pd=0 nrd=0 nrs=0 par=1 dtemp=0 sa=0 sb=0 sd=0
46946 .param par_vth=0.00666
46947 .param par_k=0.002833
46948 .param par_l=1.5e-7
46949 .param par_w=-1e-7
46950 .param par_leff='l-par_l'
46951 .param par_weff='par*(w-par_w)'
46952 .param p_sqrtarea='sqrt((par_leff)*(par_weff))'
46953 .param var_k='0.7071*par_k*1e-06/p_sqrtarea'
46954 .param mis_k='agauss(0,var_k,1)'
46955 .param var_vth='0.7071*par_vth*1e-06/p_sqrtarea'
46956 .param mis_vth='agauss(0,var_vth,1)'
46957 m0 d g s b pfet_03v3 ad=ad as=as delvto='mis_vth*sw_stat_mismatch' l=l m=m mulu0='1-mis_k*sw_stat_mismatch' nf=nf nrd=nrd nrs=nrs pd=pd ps=ps sa=sa sb=sb sd=sd w=w
46958 .ends pfet_03v3

```

$$PAR_{V_{TH}}=0.007148, \text{ var_vth} = \sigma_{\Delta V_{TH}} = \frac{1}{\sqrt{2}} \frac{PAR_{V_{TH}}}{\sqrt{WL}} = 5.054 \text{ mV}/\mu\text{m}$$

As $PAR_{V_{TH}}$ is for pair mismatch of two MOSfets.

Upon comparing the value from the PDK and from the SA (ADT) we found that the value of the $\sigma_{\Delta V_{TH}}$ in the PDK file is lower and this is due to the fact that it is not the $\sigma_{\Delta V_{TH}}$ only but is $\sigma_{\Delta V_G}$ which takes into account the change of K constant which includes mobility and effective dimensions variations so it is reasonable for

$$V_{gmis} \text{ be greater than } \sigma_{\Delta V_{TH}} \text{ that we found in the PDK as } \sigma_{V_g} = \sqrt{\sigma_{V_{TH}}^2 + \left(\frac{V_{OV}}{2} \frac{\sigma_K}{K}\right)^2}$$



Examine these trade-offs using SA

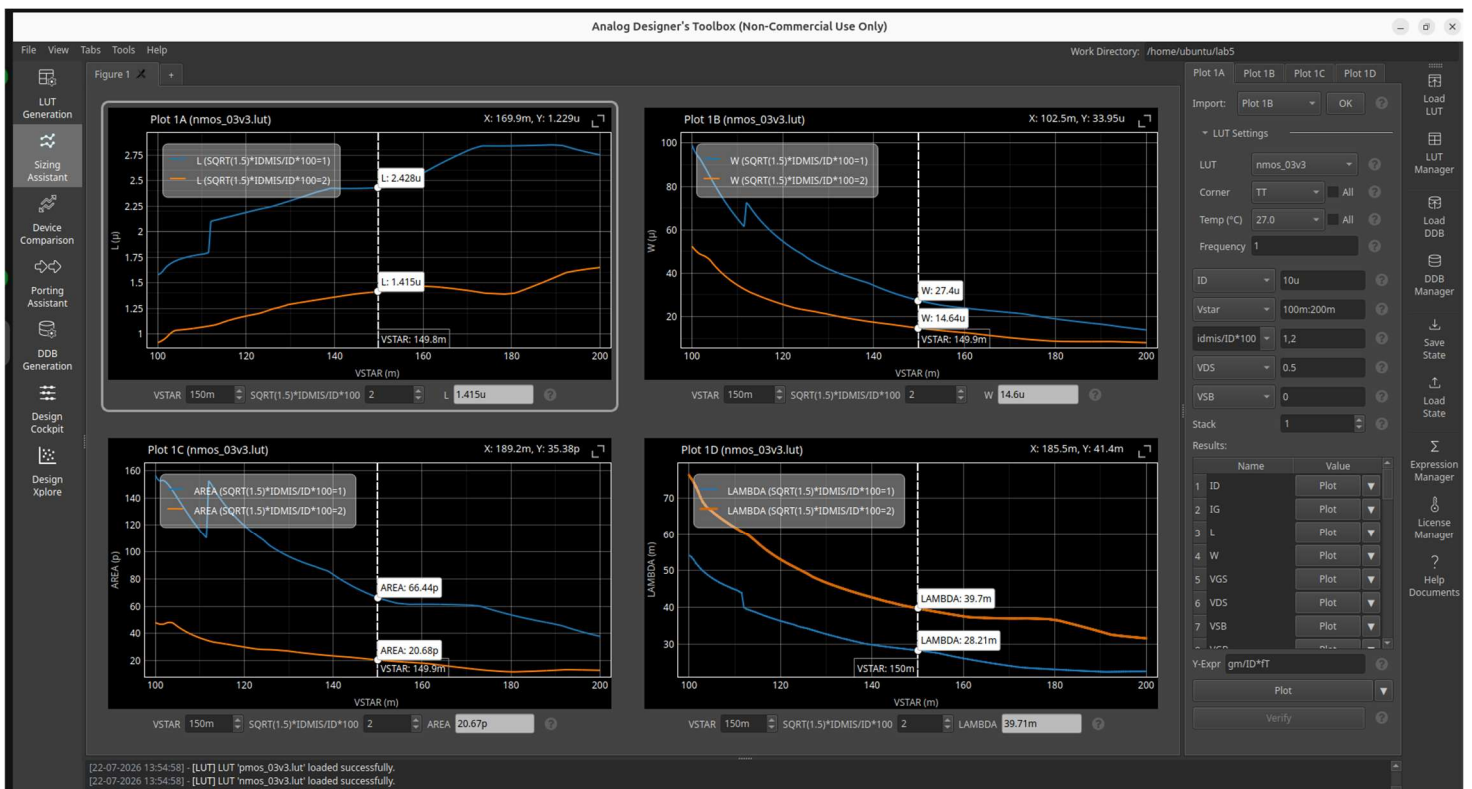
$$\frac{\sigma(I_{out})}{I_{out}} = \frac{\sqrt{1 + \frac{1}{m}} i_{d,mis}}{I_D} \times 100$$

Which is equivalent to

$$\frac{\sigma(I_{out})}{I_{out}} = \frac{2}{V^*} \frac{A_{VT}}{\sqrt{LW}} \sqrt{1 + \frac{1}{2}} \times 100 \leq 2$$

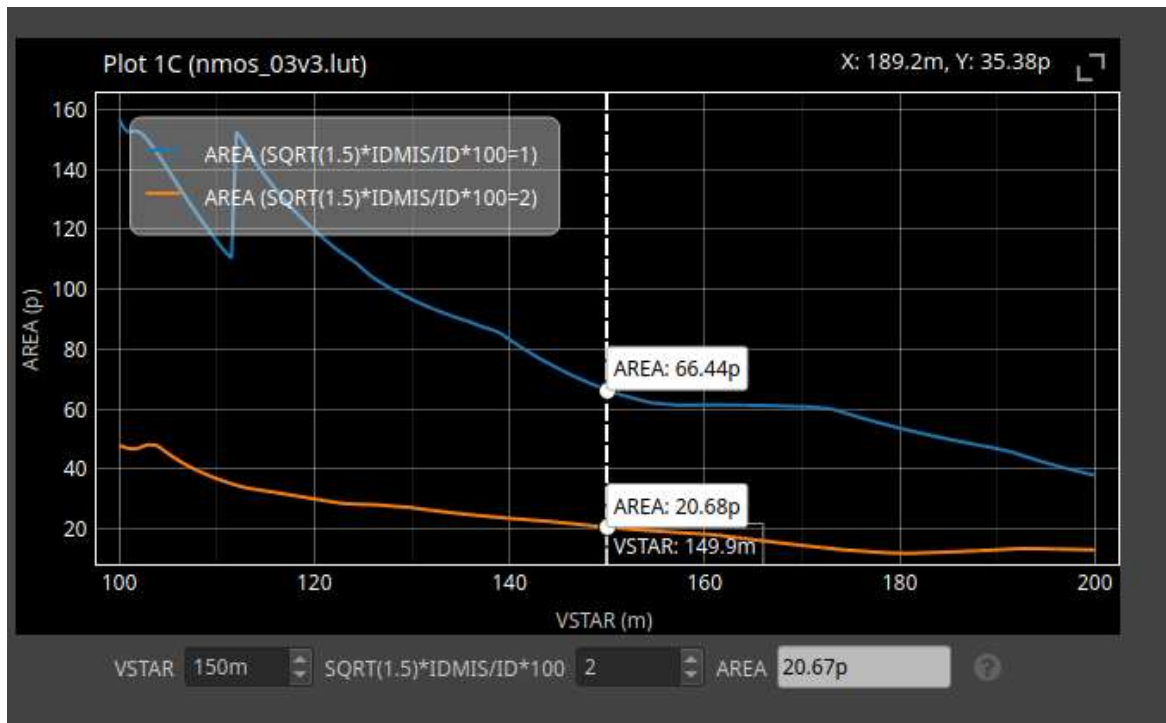
$$LW \geq \left(\frac{A_{VT}}{V^*} \sqrt{\frac{3}{2}} \times 100 \right)^2$$

So we want our area to be less than the expression above



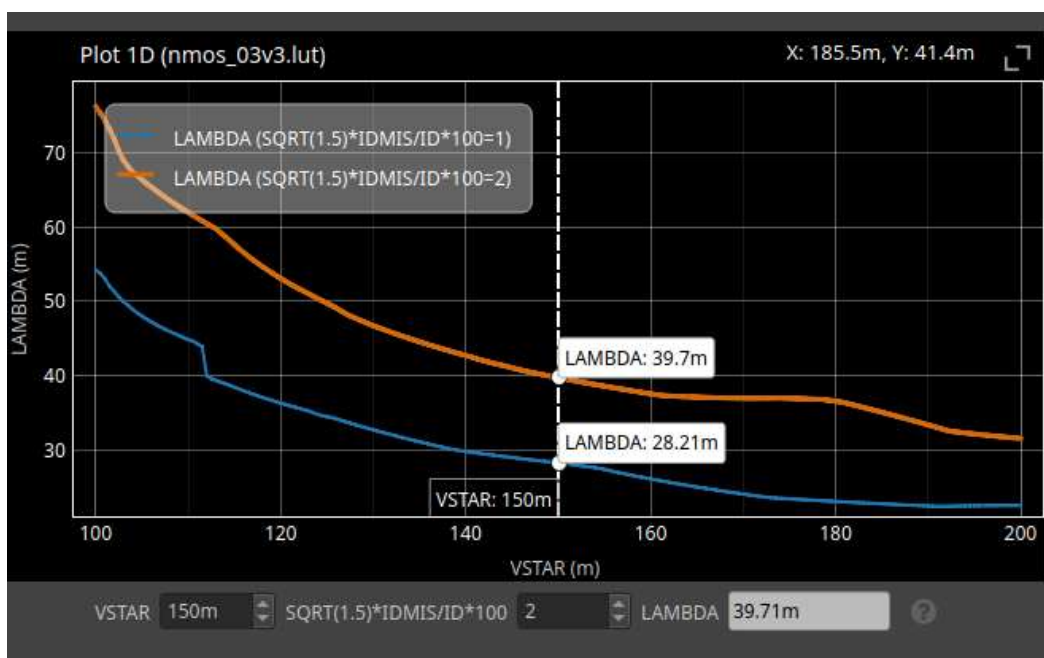


Area vs V*



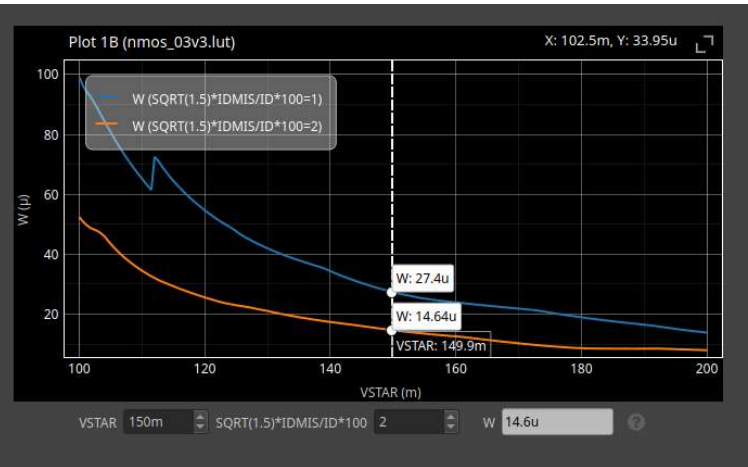
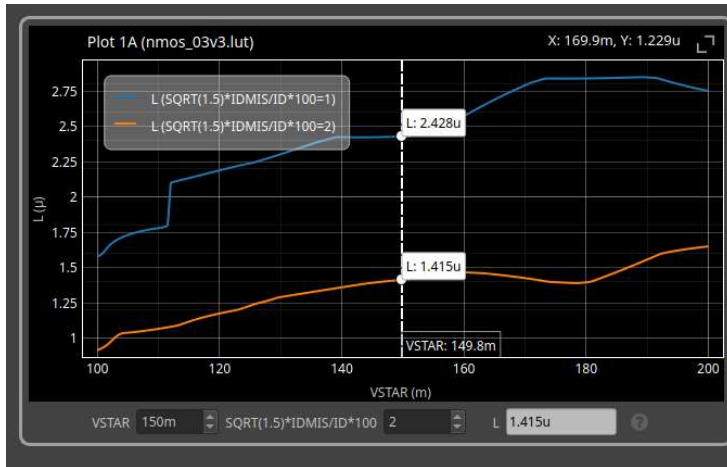
So, we want the maximum V* star available and we will use an $\frac{\sigma(I_{out})}{I_{out}} = 2\%$

Check if it satisfies the Lambda requirement





L and W specs



$L = 14.6\mu\text{m}$, $w = 1.42\mu\text{m}$

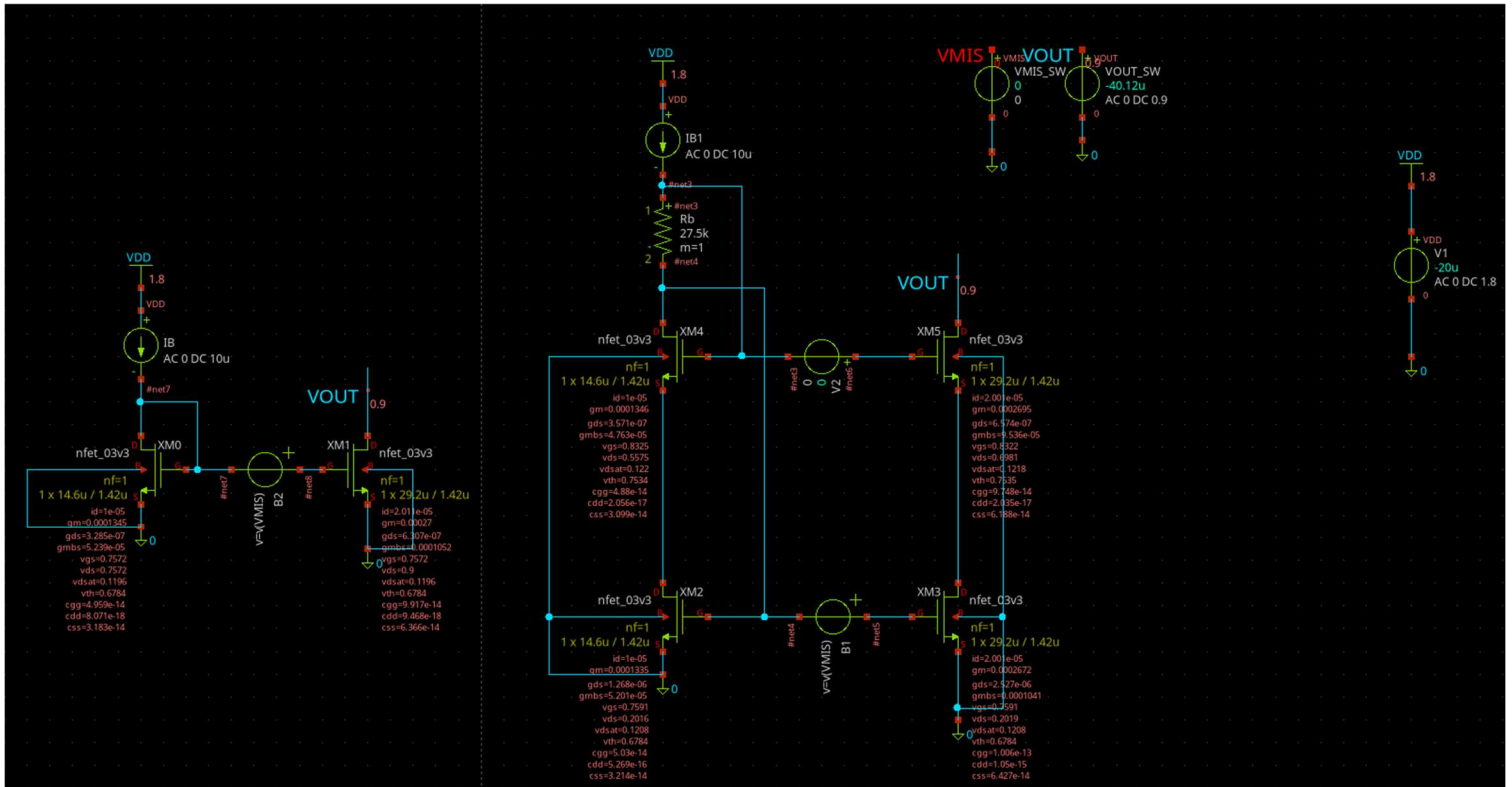
Check for satisfactory of the required specs

Yes it satisfies the lambda at 39.7 m and the $\frac{\sigma(I_{out})}{I_{out}}$ is also satisfied at 2%



Part 2: Current Mirror Simulation

Schematic



Design and OP (Operating Point) Analysis

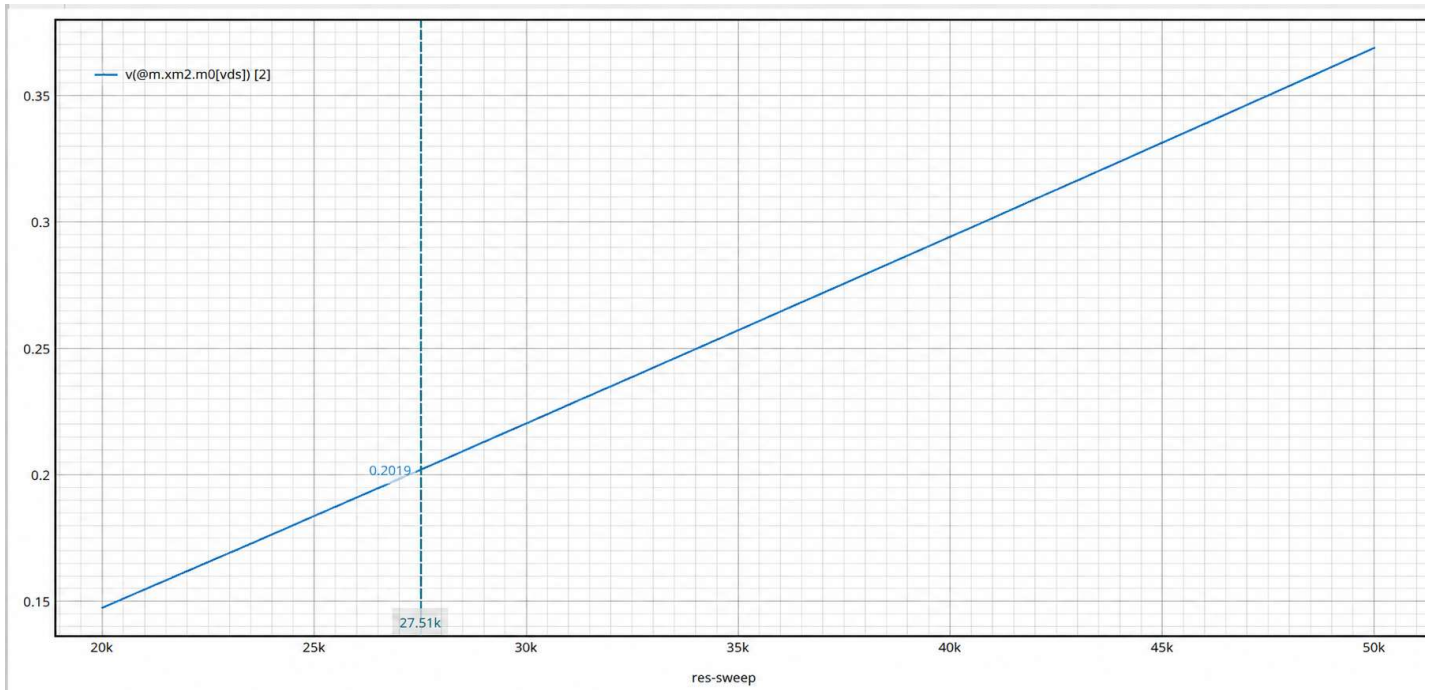
Rough Calculation of R_B

$$R_B = \frac{V_{GS} + V_{DS} - V_{GS}}{I_B} \approx \frac{V_{DS}}{I_B}$$

So assuming $V_{DS2} = V^* + 50\text{mv} = 200\text{mv}$ then $R_B = 20k\Omega$



DC Sweep for V_{DS2} against R_B

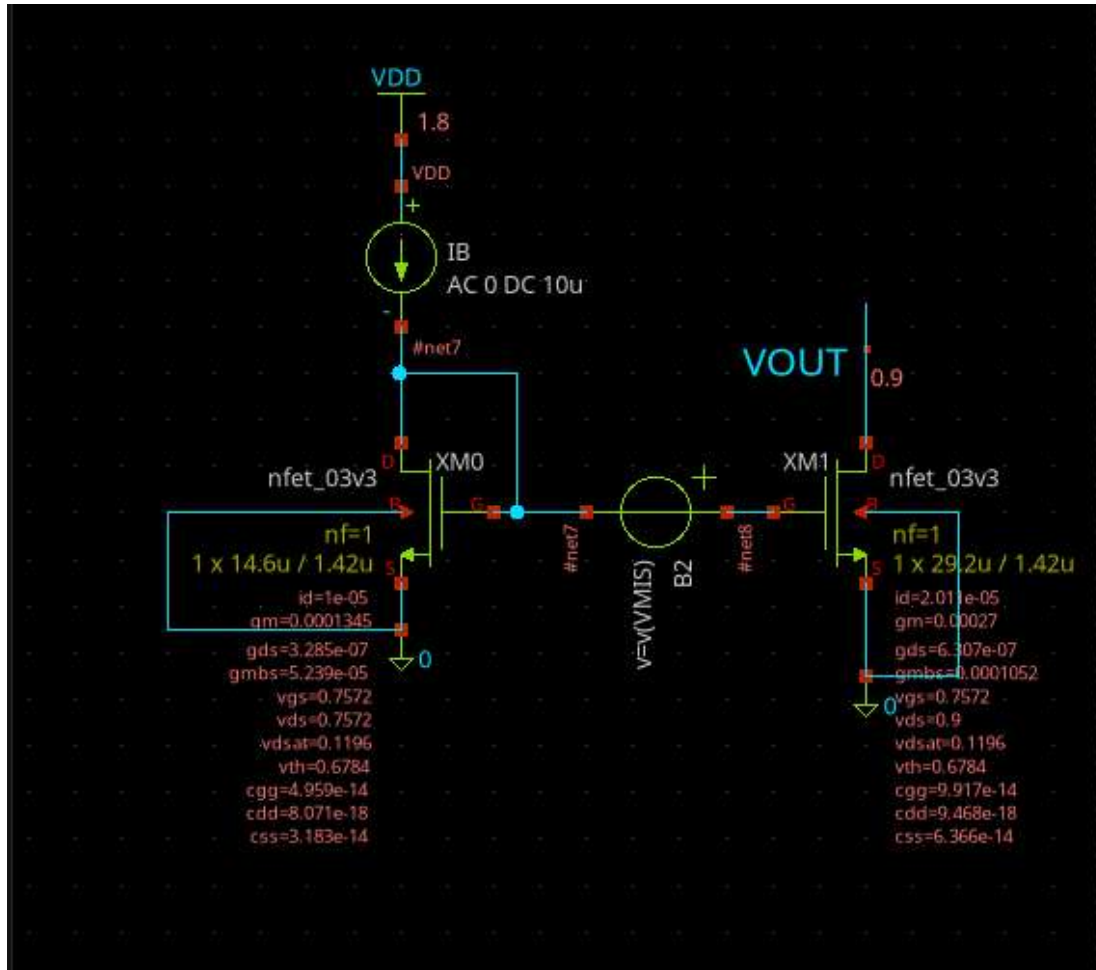


We will find the suitable R_B which achieves our V_{DS} specs at 27.5k Ω which is higher than the 20k Ω expected and this due to the V_{th4} being higher than that of V_{th2} so they won't have equal V_{gs} due to body effect .



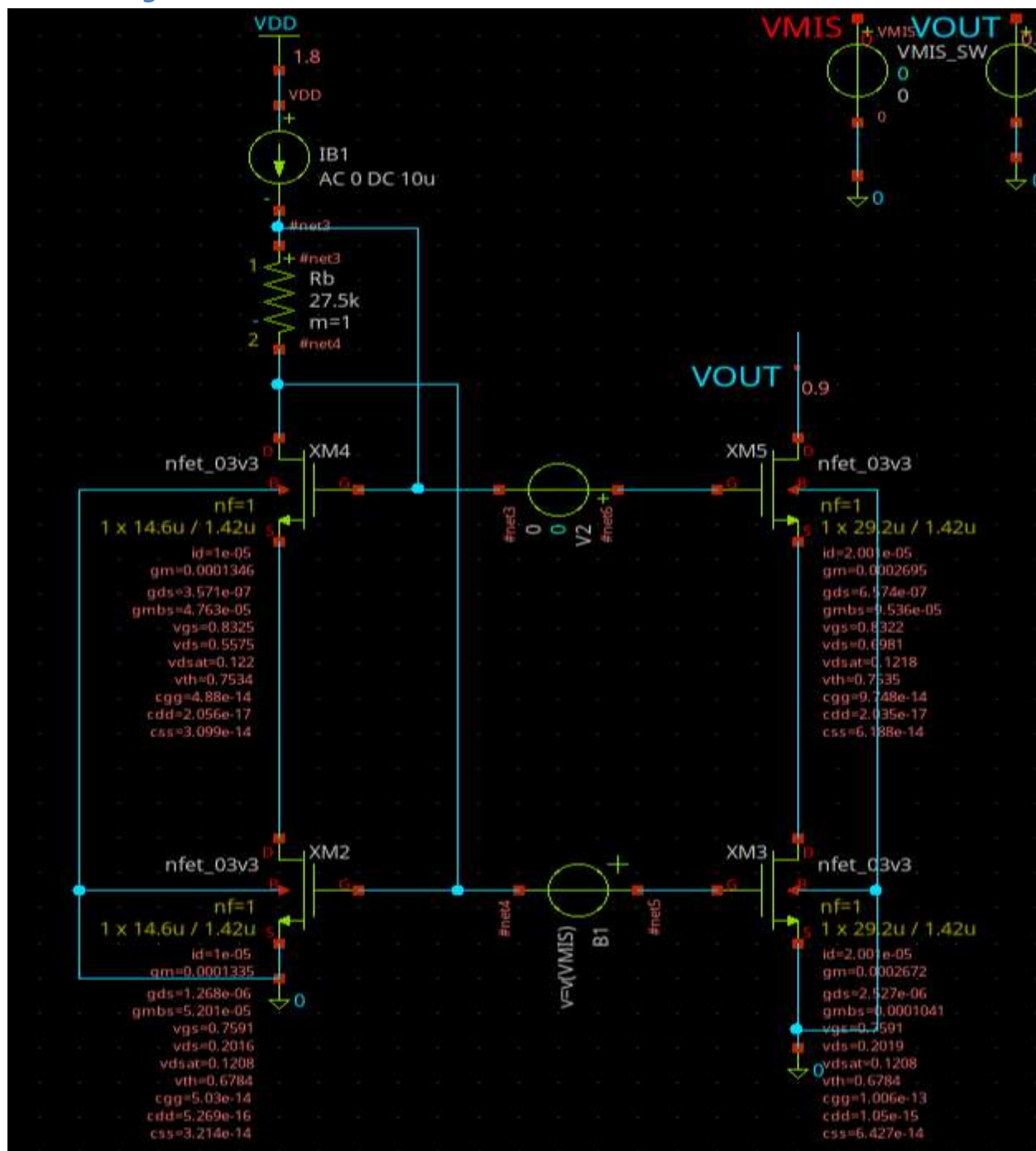
Simulate the OP point

Simple current mirror





Wide Swing current mirror

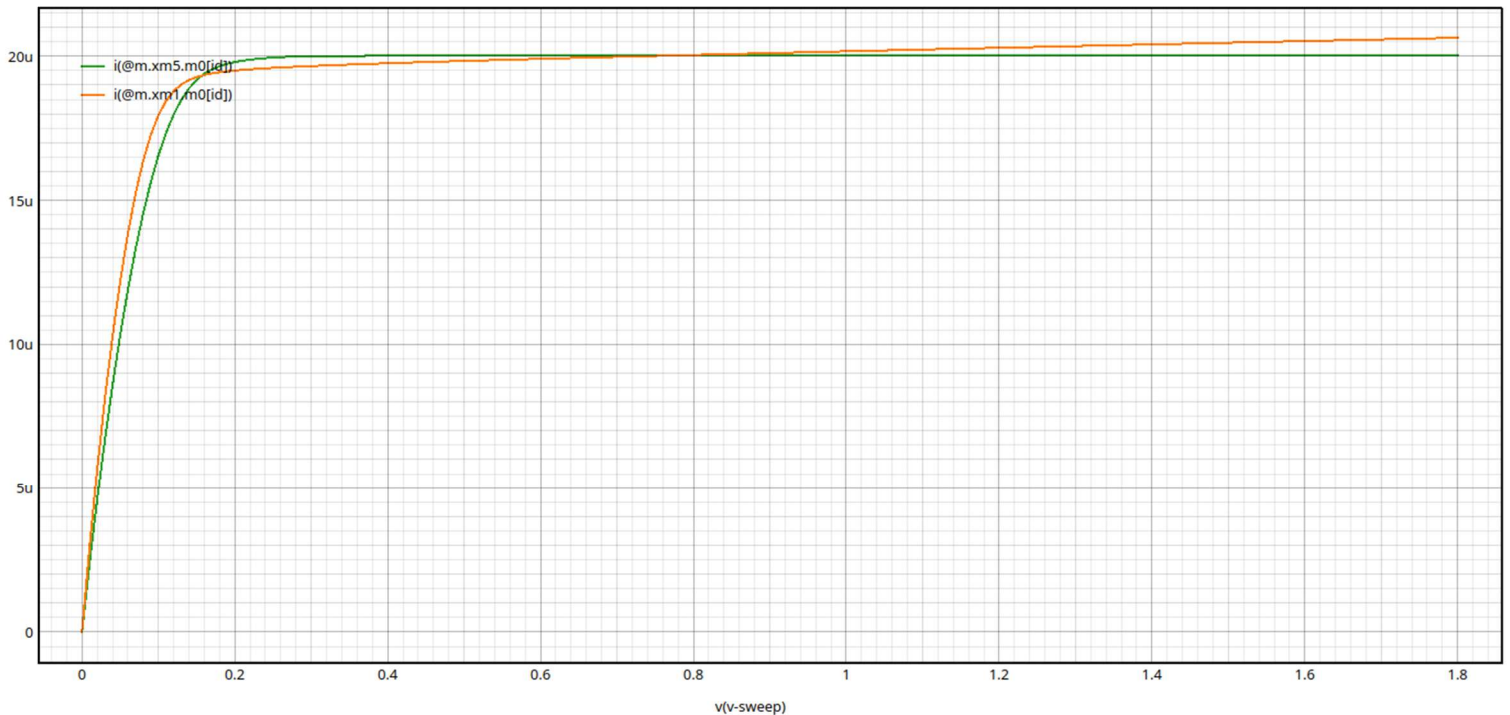


Mosfets operation in saturation check

Yes all mosfets operate in saturation as the V_{DS} is greater than V_{Dsat}



DC Sweep (I_{out} vs V_{OUT})



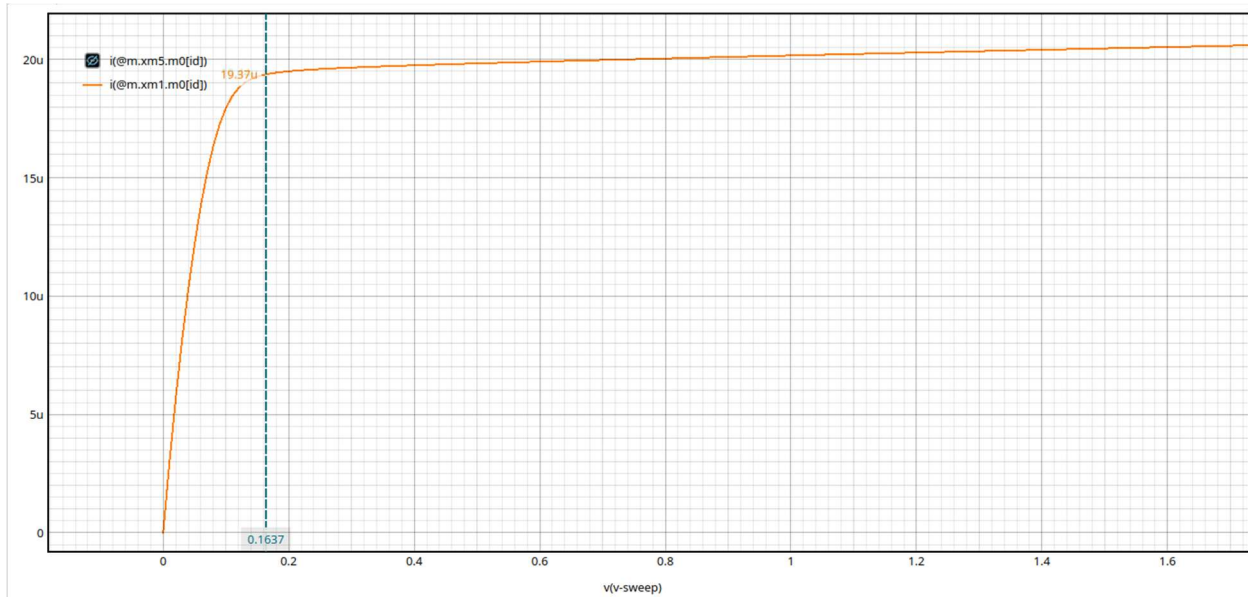
Comment:

The normal current mirror is more sensitive to the variation in v_{out} as the dependency on V_{ds} is higher due to the V_{ds} of the two Mosfets are not equal and one could change without being related to the other side which causes this stronger variation in current while the cascoded current mirror the V_{ds} changes less aggressively on the mirror device which ensures constant current as the Drain node for the current mirror device is almost constant with variation around the 18mV and this won't change the current considering that it is V_{DS} which is a weak voltage with a minimal change on current.



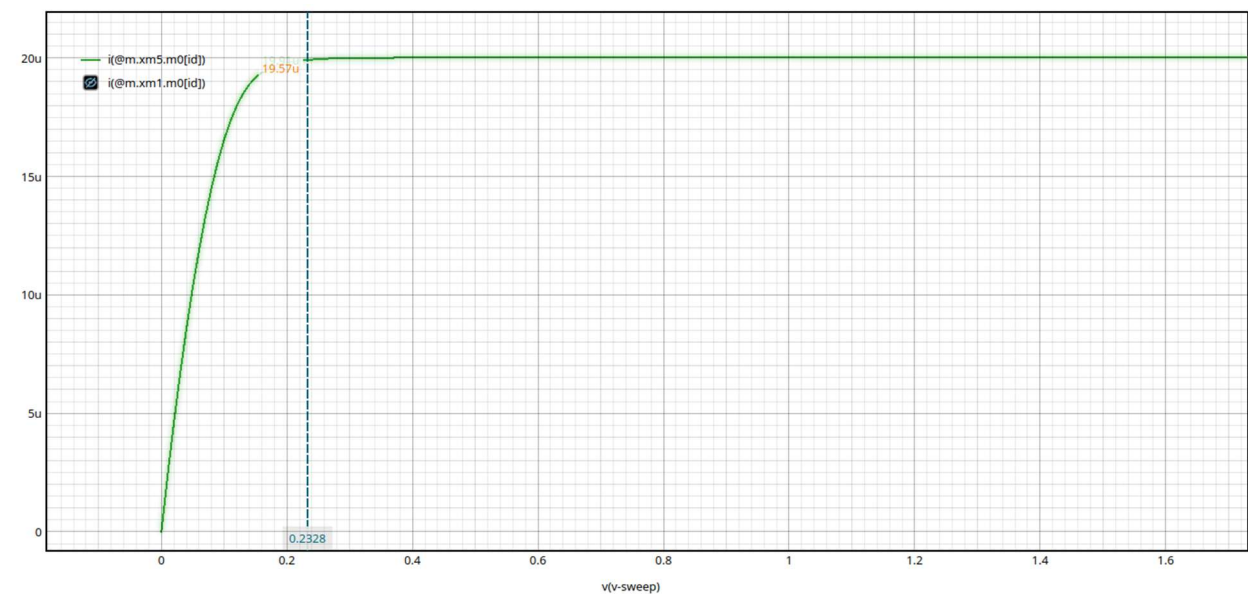
Compliance voltage of each circuit from graph

Simple current mirror



Around 163 mV which is close to analytical results of 150mV theoretically it should be 120mV as V_{Dsat} is 120mV but due to being at the edge of saturation so you will have a small r_o thus the v_{DS} will have a large effect on current so we can't consider it the actual compliance voltage.

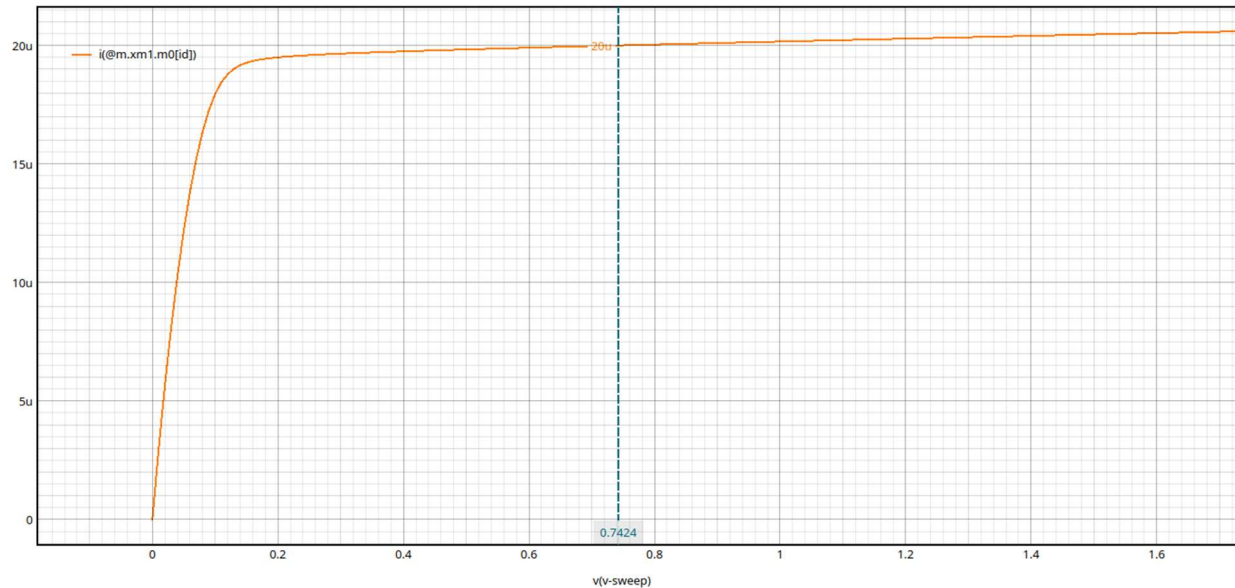
Wide swing current mirror





Around 233mV which is also reasonable and here it is clear that it was more near the anticipated 250mV as the $V_B=1\text{v}$ approx. and $V_{th5}=0.75\text{ V}$ so to be in saturation for this mosfet we need V_{out} around 250mv which is close to 233mV

I_{out} of the simple CM is exactly equal to I_B*2 at a specific value of V_{OUT}



As seen here when V_{out} is around 742mV the $I_{out} = 2 * I_B$ and this due to at this exact value the V_{DS} of the two mosfets become equal.



Calculate the percent change in I_{out} when V_{OUT} changes from 0.5V to 1.5V

```
Using SPARSE 1.3 as Direct Linear Solver
No. of Data Rows : 181
Simple Current Mirror Simulation Results
i1          = 1.983370e-05
i2          = 2.046595e-05
perc_change = 3.137744e+00
Wide Swing Current Mirror Simulation Results
i1          = 2.001215e-05
i2          = 2.001680e-05
perc_change = 2.323318e-02
binary raw file "cmirror dc.raw"
binary raw file "/home/ubuntu/lab5/simulation/untitled.sch/untitled.raw"
Doing analysis at TEMP = 27.000000 and TNOM = 27.000000

Using SPARSE 1.3 as Direct Linear Solver

Total analysis time (seconds) = 0.00461234

Total elapsed time (seconds) = 0.028

Total DRAM available = 7376.586 MB.
DRAM currently available = 3998.453 MB.
Maximum ngspice program size = 83.457 MB.
Current ngspice program size = 19.629 MB.

Shared ngspice pages = 12.488 MB.
Text (code) pages = 6.406 MB.
Stack = 0 bytes.
```

Simple current mirror change

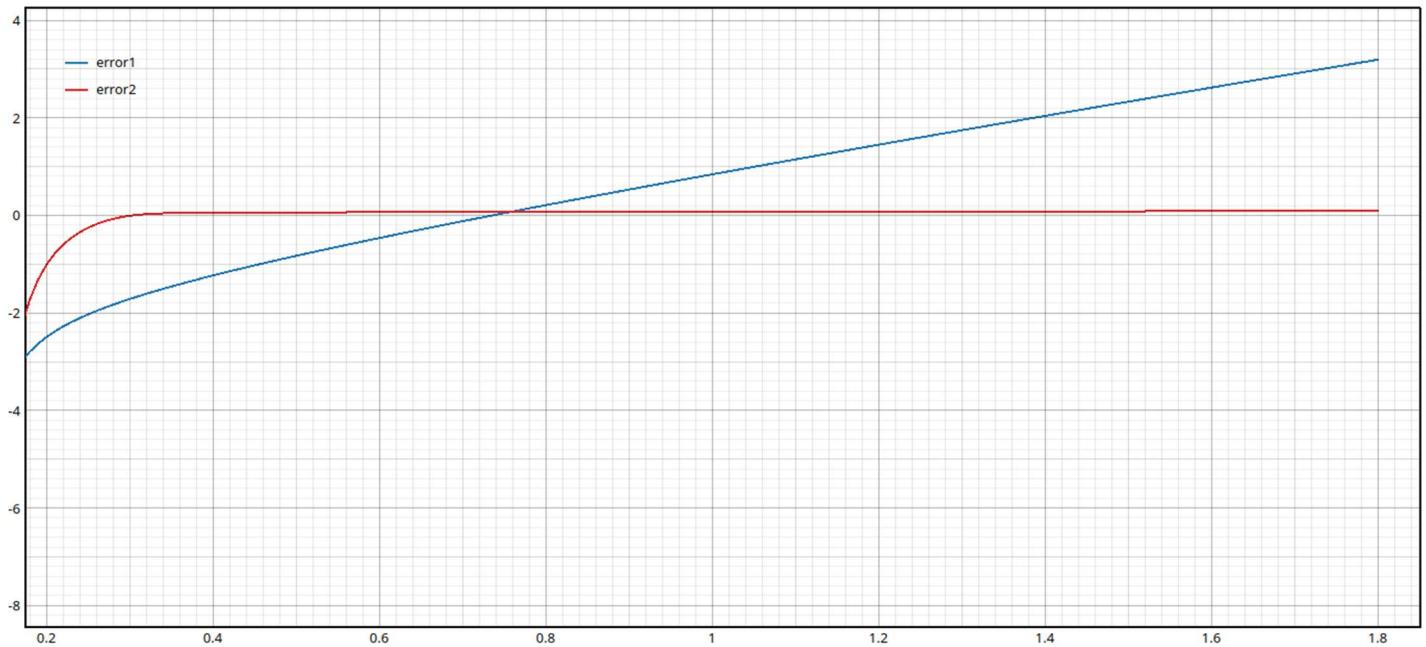
Is 3.13774% comparing with the value we obtained before it is less than maximum change of 10% and λ from SA is 39.7m and from dc analysis is 31m due to higher V_{DS} linearizing around 31m for DC analysis gives us a percentage of $1 \cdot (31e-3) \cdot 100 = 3.1\%$ which is pretty close and if use the value from SA it will be 3.9% which is still acceptable margin of error due to the change in V_{DS}

Comment on the difference between the two circuits

It is obvious that the WS has less change thus better CS characteristic due to the holding the value of V_{DS} unlike the other in which the V_{out} affects directly V_{DS}



The percent of error in I_{out} vs V_{OUT}

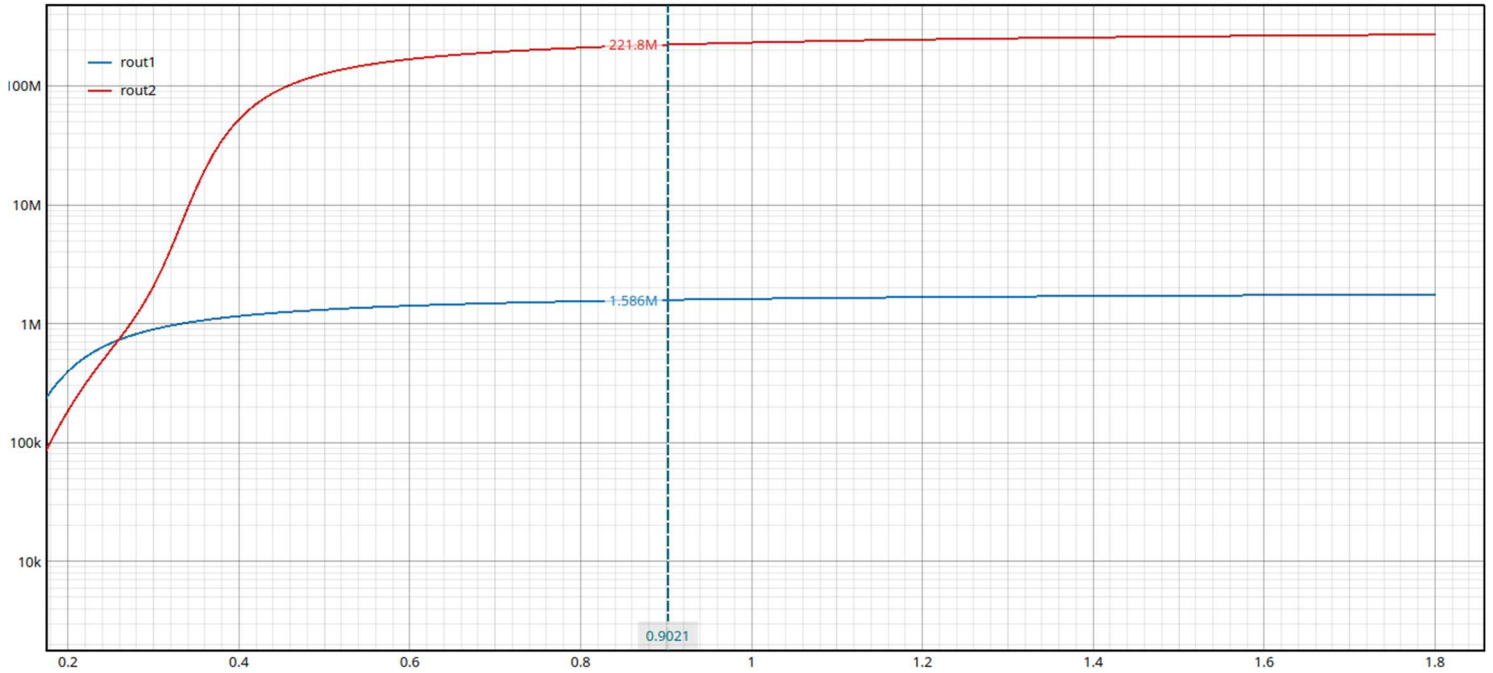


Comment:

It's clear that the WS has less error and more stable error and this is again due to the more stable V_{DS} while the simple current mirror V_{DS} is strongly affected by the change in V_{out} which cause a less stable I_{out} and more error.



Rout vs VOUT



Comment:

It is clear that the Rout of the cascaded wide swing CM is larger as the cascode topology known for boosting the Rout .

Effect of Vout on Rout

Rout changes with vout as vout affects the V_{DS} and V_{DS} affects directly V_a increasing r_o which increases Rout but reaches a certain value after which it saturates

Analytical calculations of Rout

$$\text{Simple CM: } R_{out,1} = r_{o1} = \frac{1}{g_{ds1}} = \frac{1}{6.007 \times 10^{-7}} = \boxed{1.665 \text{ M}\Omega}$$

$$\text{Wide Swing CM: } R_{out,2} = r_{o5} [1 + (g_{m5} + g_{mb}) R_s]$$

$$\boxed{R_{out,2} \approx 221.15 \text{ M}\Omega}$$

Topology	Analytical(R_{out})	Simulated(R_{out})	Error
Simple current mirror	(1.665MΩ)	(1.586MΩ)	(4.96%)
Cascode current mirror	(221.15MΩ)	(221.8MΩ)	(0.29%)



Mismatch

DC Sweep for VMIS_SW

```

No. of Data Rows : 15
Simple Current Mirror Simulation Results
i1          = 2.010552e-05 at= 0.000000e+00
i2          = 2.048592e-05 at= 1.400000e-03
perc_change = 1.874287e+00
Wide Swing Current Mirror Simulation Results
i1          = 2.001434e-05 at= 0.000000e+00
i2          = 2.038811e-05 at= 1.400000e-03
perc_change = 1.850234e+00
binary raw file "cmirror_dc.raw"
Doing analysis at TEMP = 27.000000 and TNOM = 27.000000

Using SPARSE 1.3 as Direct Linear Solver

No. of Data Rows : 15
Wide Swing Current Mirror Simulation Results (Cascode Mismatch)
i1          = 2.001434e-05 at= 0.000000e+00
i2          = 2.001691e-05 at= 1.400000e-03
perc_change = 1.283997e-02
binary raw file "cmirror_dc.raw"
binary raw file "/home/ubuntu/lab5/simulation/untitled.sch/untitled.raw"
Doing analysis at TEMP = 27.000000 and TNOM = 27.000000

Using SPARSE 1.3 as Direct Linear Solver

Total analysis time (seconds) = 0.00166058

```

Analytical change

Simple CM

From the PDK $\sigma_{VT} = 5.05$ mv for unity area for our area it is equal 1.1mv
 so $\sigma_{VTtotal} = 1.36$ mv then $\sigma_{Iout} = gm * \sigma_{VTtotal} = 0.00027 * 1.36$ m=3.672e-7

$$\text{Percentage} = \frac{3.672e-7}{20e-6} * 100 = \boxed{1.836\%}$$

Wide Swing CM

Same but there is only a slight difference in gm=0.0002672s

So the $\sigma_{Iout} = 3.63392e - 7$

$$\text{Percentage} = \frac{3.634e-7}{20e-6} * 100 = \boxed{1.817\%}$$

The results are very close the simulation results



Cascode mismatch

```

No. of Data Rows : 15
Wide Swing Current Mirror Simulation Results (Cascode Mismatch)
i1      = 2.001434e-05 at= 0.000000e+00
i2      = 2.001691e-05 at= 1.400000e-03

```

Analytical

We will treat the mismatch voltage as a small signal and the G_m will be the degenerated $G_m = \frac{g_m}{1+(g_m+g_{mb})R_S} = 1.850 \mu S$

$$\sigma_{I_{out}} = 1.850 \mu * 1.36 m = 2.516e-9$$

Which mismatch is more pronounced

The current mirror devices is more pronounced than the cascode devices mismatch as the G_m for the current mirror device is larger than that when the mismatch is coming from the cascode as the cascode G_m can be treated as the degenerated g_m of a common source due the device below it .

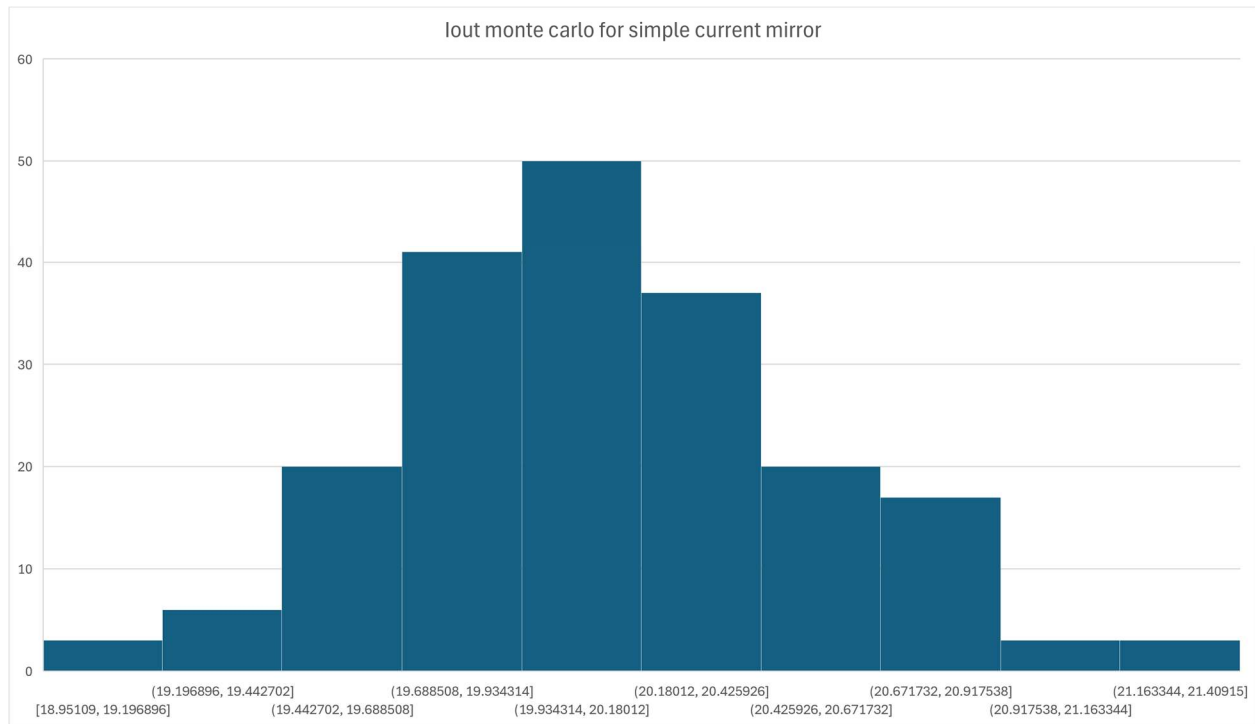
Area design choice

From what we have seen it is clear that having a larger area for the current mirror devices is better as increasing the area decreases the mismatch per unit area's rule and it is clear that the mismatch in the mirror device causes more change in current due it having the bigger G_m .



Monte Carlo Simulation

Simple current mirror



Using excel to calculate the standard deviation which is equal to

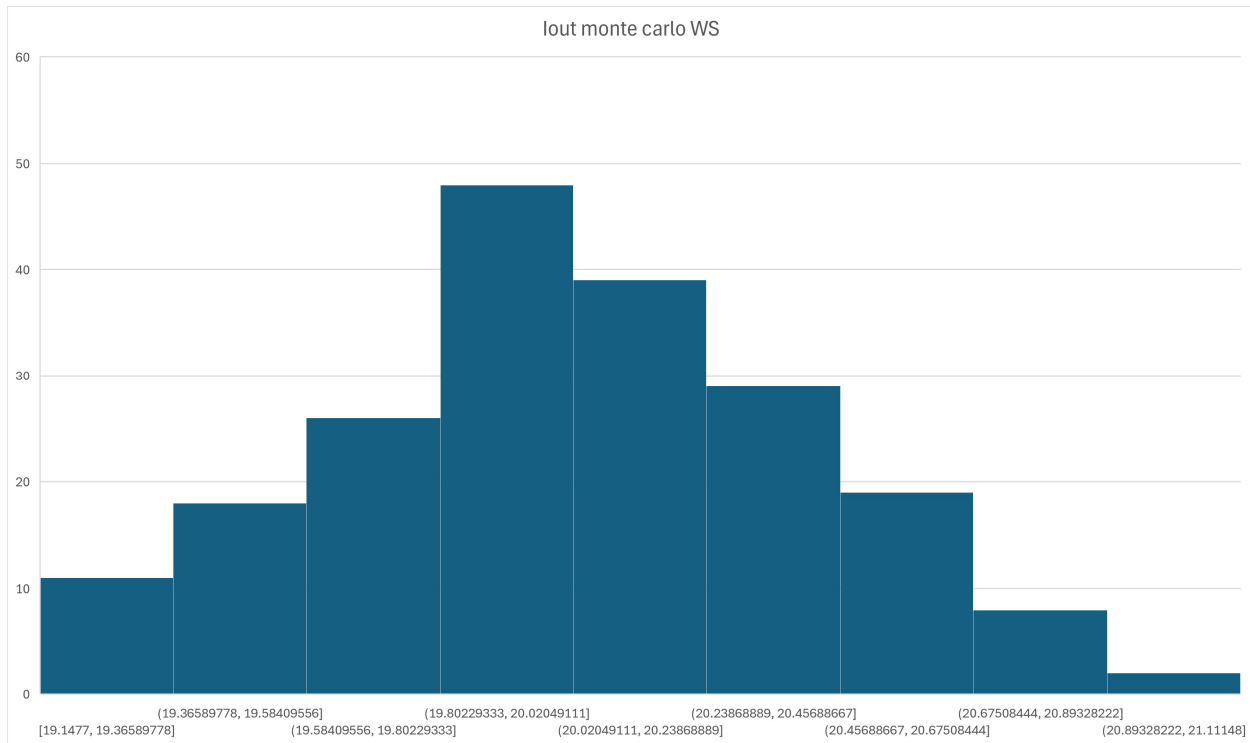
σ_{lout}
0.423697

$$\frac{\sigma(I_{out})}{I_{out}} = \frac{0.4237}{20} = 2.1185\%$$

Which is comparable to the analytical of 1.836%



Wide Swing current mirror



Using excel to calculate the standard deviation which is equal to

σ_{lout}
0.395075

$$\frac{\sigma(I_{out})}{I_{out}} = \frac{0.395075}{20} = 1.9753\%$$

Which is comparable to the analytical of 1.817%